```
 #include<iostream>
using namespace std;

class node
{
    public:
    int data;
    node*next;
    node(int x)
    {
        data=x;
        next=NULL;
    }
};
class queue
{
    private:
    node*front;
    node*rear;
    public:
    queue()
    {
        front=NULL;
    }
    int empty()
    {
        if(front==NULL)
        return(1);
        else
        return(0);
    }
    void enqueue(int x)
    {
```

```
    node*q=new node(x);
    if(empty())
    {
        front=q;
        rear=q;
    }
    else
    {
        rear->next=q;
        rear=q;
    }
}
int dequeue()
{
    int x=front->data;
    node*q=front;
    front =front->next;
    delete q;
    return(x);
}
int peek()
{
    return(front->data);
}
void display()
{
    node*q=front;
    while(q!=NULL)
    {
        cout<<"\t"<<q->data;
        q=q->next;
    }
}
```

```
};

int main()
{
    queue q;
    int x;
    q.enqueue(10);
    q.enqueue(20);
    q.enqueue(30);
    q.display();

    x=q.peek();
    cout<<"front element is"<<x<<endl;
    x=q.dequeue();
    cout<<"removed"<<x<<endl;
    q.display();

    q.dequeue();
    q.dequeue();
     q.dequeue();  //output : stack underflow
    return(0);

}

//output

/*PS D:\kanchankharat>> cd "d:\kanchankharat\" ; if ($?) { g++ queue_using_link
edlist.cpp -o queue_using_linkedlist } ; if ($?) { .\queue_using_linkedlist }
        10      20      30front element is10
removed10
        20      30*/
```